

# ALI NAQVI

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## EDUCATION

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### University of Waterloo

*Bachelor of Mathematics (BMath), Co-op*

Waterloo, ON

*Sept 2026 – April 2031*

## TECHNICAL SKILLS

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**Languages:** Python, Rust, C, C++, JavaScript, TypeScript, ARM64 Assembly, Cython, HTML, CSS

**Technologies:** Git, GitHub, PyTorch, Hugging Face, FastAPI, Docker, TensorFlow, Keras, Flask, React

## EXPERIENCE

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### EPSi, Florida International University

*Undergraduate Research Intern*

July 2026 – August 2026

*Miami, FL (Remote)*

- Built a privacy-preserving Bonawitz-style **Secure Federated Learning** pipeline on a **PyTorch 1D-CNN (46k parameters)**, detecting solar inverter faults across 37 clients at **85.2% accuracy** and **88.3% F1** over 20 training rounds.
- Implemented a 4-round protocol (**ECDH P-384**, Shamir's Secret Sharing over **GF(2<sup>521-1</sup>)**, **double masking**, **ECDSA**), cutting gradient leakage similarity from 0.98 to **0.04** with **0.0 error** in aggregate reconstruction.
- Withstood **30% client dropout** without aggregate corruption; profiled edge deployment at **3.84s secure aggregation** and **0.22ms** single-core inference.

### St. Joseph's Healthcare Hamilton

*Machine Learning Intern*

June 2026 – August 2026

*Hamilton, ON*

- Fine-tuned **Llama-3-8B-Instruct** (4-bit) with **rank-16 LoRA** across **7 attention and MLP projections** via **MLX**, training **400 iterations** to a **0.265 validation loss**.
- Built a clinical triage engine on synthetic **matched triads** that separate active lung nodules from history-only findings, reaching **99.9% accuracy** with a **100% sensitivity** oncology failsafe.
- Cut false clinical alarms by over **80%** by teaching the model to distinguish **contextual nodule statuses** that bypass standard **regex pipelines**.

## PROJECTS

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### mtdi | Rust, Systems Programming, arm64 Assembly

[github.com/anaqvi02/mtdi](https://github.com/anaqvi02/mtdi)

- Built and open-sourced a **dynamic instrumentation engine** for arm64 macOS: **10–100x faster than Frida**, matching Dobby.
- Engineered inline hook detours at **1.6 ns/call** syscall and **15 ns/call** full-context cost, with live-PID injection via **task\_for\_pid** and remote thread bootstrapping.
- Designed a probe compiler with an **AST verifier** enforcing zero-panic rules, feeding an allocation-free **SPSC ring buffer** for lock-free telemetry.

### PicoEmulator | Python, Cython, ARM Architecture

[github.com/anaqvi02/PicoEmulator6000](https://github.com/anaqvi02/PicoEmulator6000)

- Built an **ARM Cortex-M0+ emulator** in **Python** simulating the RP2040 (Raspberry Pi Pico), implementing the full **Thumb-1 instruction set**.
- Engineered the core in **Cython** for a **1000x speedup** over pure Python, using **direct memory access** and inline fast-paths to hit ~33 MHz.
- Added **benchmarking tools** measuring performance across technical workloads.